

The sensor distinguishes itself with a hardware-centric master/slave I<sup>2</sup>C interface that integrates seamlessly with software-emulated universal



asynchronous receiver-transmitter (UART), serial peripheral interface (SPI), and pulse-width modulation (PWM) interfaces, setting it

as a top contender for sub-1GHz applications and offering scalability for Bluetooth Low Energy (BLE) TPMS. Upholding the automotive safety integrity level (ASIL), it embodies a high level of integration and is fine-tuned to encompass all the vital functionalities required in a modern TPMS module. It is suitable for battery-operated applications.

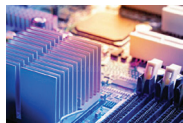
Infineon Technologies

<https://www.infineon.com>

## TEST & MEASUREMENT

### Next-generation protocol analysis testing platform

VIAVI Solutions Inc. is launching a new protocol analysis testing platform, namely PCIe 6.0, at flash memory summit for various industries. PCI Express (PCIe) is a swift computer expansion bus standard, fostering emerging applications across data centres, cloud, AI,



quantum computing, IoT, automotive, and defence industry sectors. Running at 64GT/s data rates,

PCIe 6.0 elevates bandwidth and throughput twofold compared to the preceding PCIe 5.0 specification. This extends the equipment's lifespan during the initial development R&D stage. The upcoming Xgig family will feature a PCIe 6.0 protocol analysis testing platform encompassing analysers, exercisers, and interposers. This innovative platform is purpose-built to offer profound insights and will encompass the familiar analysis capabilities of the Xgig suite. The upcoming PCIe 6.0

platform will encompass well-known Xgig analytical functionalities such as Xgig software suite, protocol support, link training and status state machine (LTSSM) override, auto tuning and port bifurcation.

VIAVI Solutions Inc.

<https://www.viavisolutions.com/en-us>

### Electronics inspection with new AOI system

Test Research Inc (TRI) has launched the AOI TR7700 SIII Ultra, an advanced high-volume automated optical inspection (AOI). Catering to the evolving demands of the modern industrial landscape, it seamlessly



integrates with Industry 4.0 protocols, aligning with contemporary smart factory benchmarks and rigorously adhering to the Hermes standard. With a rapid speed of 60cm<sup>2</sup>/sec, the high-speed AOI system combines AI algorithms, smart programming, and metrology measurements to ensure deep, accurate inspections. The equipment's intelligent programming ensures rapid readiness, making it operational for inspections in minutes and doing away with extended setup durations. Acknowledging the value of perpetual enhancement, the system is equipped with a closed-loop structure, integrating feedback and feedforward mechanisms. This guarantees a responsive and forward-thinking inspection process, aptly adjusting to immediate and impending needs. The smart factory adoption permits effective monitoring and control of inspection status, revealing trends and vital data, and swiftly resolving discrepancies.

Test Research Inc. (TRI)

<https://www.tri.com.tw/en>

### Radar testing systems for autonomous vehicles

Rohde & Schwarz has launched the R&S AREG800A automotive radar

echo generator and the R&S QAT100 antenna array. The system simulates comprehensive driving scenarios to rigorously test radar-based advanced driver assistance systems (ADAS) and radar sensors integrated in automated driving (AD) cars, fully over the air. This upgrade heralds a future where



autonomous braking tests reach unmatched realism, achieved by generating exceedingly short distances for artificial objects. The revamped radar test system (RTS) emerges as the tool for radar object simulation, encapsulating single sensor trials to immersive 360° environments, and specifically catering to the 76-81GHz bandwidth pivotal for automotive radar sensors. The RTS promises scalability and embodies adaptability, facilitating a vast array of electronically generated ADAS traffic scenarios throughout the testing lifecycle.

Rohde & Schwarz

<https://www.rohde-schwarz.com>

## BOARDS

### A compact microcontroller for IoT and AI integration

Adafruit has launched Metro ESP32-S3 with 16MB flash and an 8MB pseudo-static random access memory (PSRAM) development board. With a native universal serial bus (USB) and an



abundance of PSRAM, the board is suited for CircuitPython or Arduino, allowing for affordable Wi-Fi inte-

gration while maintaining compatibility with shields. The board is both ESP-IDF and Arduino compatible. Given its superior power and radio frequency (RF) capabilities, this SoC is a prime selection for diverse applications like the Internet of Things (IoT), wearable tech-